

Supplementary Material: Task-Relevant Ultrasonic Information Acquisition for Impacted Composites

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1 S1. Evidence authority and statistical units

The supplement exposes detailed controls, sensitivities, implementation diagnostics, and machine-readable provenance omitted from the compressed main paper. Numerical authority remains the 39-row canonical metrics file. Physical specimens are the first statistical unit and the six held-out experimental domains are equally weighted. State, checkpoint, candidate, seed, and learner- pair rows are repeated computational records rather than independent samples. All teacher, oracle, and component-substitution quantities are retrospective unless an evidence record explicitly marks them deployable.

The codebase refers to the supervised state-conditioned closed-loop implementation as MAVIS. This is one implementation of State-Conditioned Task-Oriented Acquisition, not the paper-level method. The code identifier `mvd_m1_o2` denotes the static deployable reference used in the final comparison; it is not a published competing method.

Record group	Content
S01–S04	Full domain and checkpoint records for spatial, sparse, static, and conditional-value comparisons.
S05–S09	Real-state, acquired-position/history, field-content (registered reconstruction), shuffled-content, and dynamic-valuation controls.
S10–S12	Component substitutions and bounded reachable-set planning records.
S13–S16	Learner-pair value stability, feedback strata, and implementation diagnostics.
S17–S18	Source hashes, chronology, and external-data feasibility boundaries.

2 S2. Spatial information and sparse recoverability

The registered main comparison uses the matched B-family scalar-to-spatial path. The distinct independent internal-field estimator (`U1_INDEPENDENT_FIELD_SENSITIVITY`) had point MAE 0.08964 and is retained only as a sensitivity path. It is not substituted for the matched 32.1% effect.

The sparse condition observes a registered 25% normalized-raster pattern and uses bilinear interpolation before the frozen field pathway. Its 89.9% retention is referenced to the matched full-field gain. The residual sparse-to- full MAE gap remains positive in all domains, and the sampling fraction has no scanner-time interpretation.

3 S3. Task-conditioned and predictor-conditioned value

The cross-objective oracle result is bounded by the learned global-mask test. Source-trained global mechanics and reconstruction masks produced the predeclared support indicator 0; the learned global mechanics mask therefore did not reproduce the oracle separation. In the reviewer-visible narrative, the latter is the field-content reference. It is operationalized by the registered normalized-RGB-MSE reconstruction objective and is not a claim about all image representations or metrics.

Under identical strict-OOF states and splits, Ridge–Huber best-action agreement was 0.675 (95% CI 0.622–0.727) (U5_RIDGE_HUBER_BEST_ACTION), and top- k agreement was 0.685 (95% CI 0.650–0.719) (U5_RIDGE_HUBER_TOPK). Ridge–MLP best-action agreement was 0.233 (95% CI 0.182–0.283) (U5_RIDGE_MLP_BEST_ACTION), and top- k agreement was 0.213 (95% CI 0.192–0.235) (U5_RIDGE_MLP_TOPK). Because the shallow MLP was substantially less accurate, these contrasts do not isolate predictor architecture at equal predictive accuracy.

4 S4. State-conditioned valuation and source controls

The registered exact-budget set regrets were 0.08171 for the static scorer (O1_STATIC_SET_REGRET), 0.07993 for the global score (O1_GLOBAL_SET_REGRET), and 0.07978 for the random median (O1_RANDOM_SET_REGRET). These source-only references annotate the static representation; they do not establish information-theoretic unobservability.

At the endpoint, the real-state trajectory recovered 26.6% (95% CI 10.6%–41.0%) of the static-to-independent-full-field error ratio (O3_FULL_FIELD_RECOVERY). Its denominator is the independent internal-field sensitivity estimator, not the matched B-family field path.

The acquired-position/history control inherits the exact positions, ordering, and costs of the recipient trajectory while removing measured content. The reviewer-visible field-content control retains the frozen **reconstruction** control mode and uses the registered image objective on the same legal trajectory. The shuffled-content control preserves recipient positions and budget while sourcing values from separated donor content. These semantics are required when interpreting the adverse real-minus-control directions.

Figure 1 provides a result-independent qualitative check across all domains. Within each sorted domain, the lexicographically first specimen was selected without using an outcome, effect size, or direction.

5 S5. Cost-constrained realization and implementation diagnostics

The reachable-set analysis is restricted to the registered two-action pool at 6.25%. The joint near-oracle compares reachable sets retrospectively; it is not deployable and does not establish arbitrary-horizon planning regret.

For A3_FEEDBACK_BENEFIT, the no-feedback reference retained an AUEBC advantage of 1.496×10^{-5} (95% CI 1.064×10^{-5} – 1.944×10^{-5}). With the registered sign convention, feedback benefit was -1.496×10^{-5} (95% CI -1.944×10^{-5} to -1.064×10^{-5}), with feedback favorable in two of six held-out domains. This is a frozen implementation diagnostic and is not generalized to other state representations or policies.

For A4_BASELINE_MINUS_MAVIS, equal-domain CAI AUEBC was 0.125053 for the supervised state-conditioned implementation and 0.124992 for the static reference. Reference minus learned was -6.114×10^{-5} (95% CI -8.461×10^{-5} to -3.777×10^{-5}), and the learned implementation

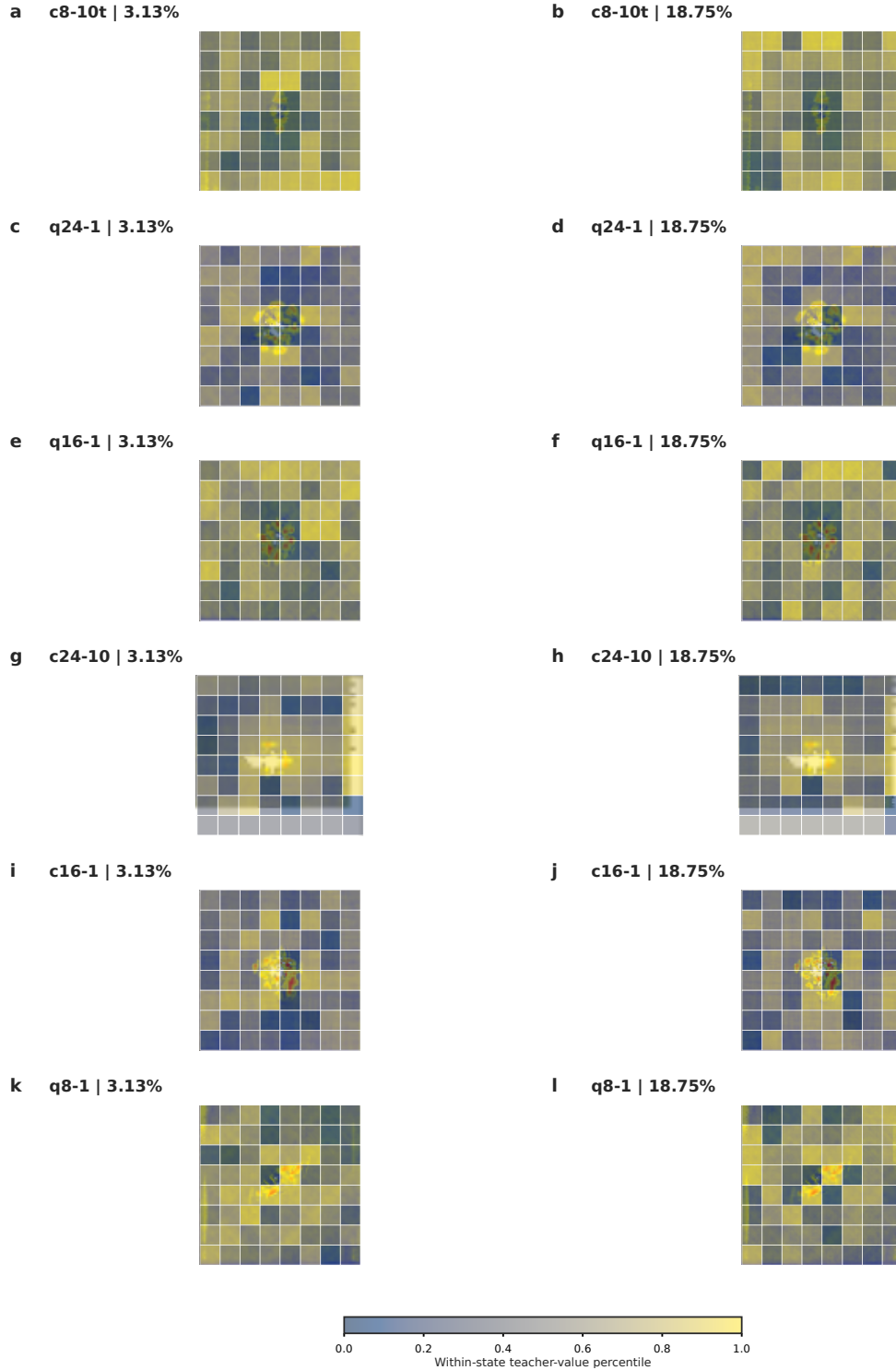


Figure 1: Cross-domain state-conditioned priority gallery. Each row shows one deterministically selected specimen from a different held-out domain. Left and right panels show the registered 3.13% initial and 18.75% states for the same trajectory. Overlays encode within-state strict-OOF teacher-value percentiles on the legal 8-by-8 action grid; they do not encode comparable absolute magnitudes across panels.

improved two of six held-out domains. The direction favors the reference. This comparison is one system-level endpoint and does not define the performance of the full Task- Relevant Information Acquisition framework.

6 S6. Provenance, chronology, and scope boundaries

The evidence chronology separates pre-existing spatial and static-reference analyses, the frozen outer implementation endpoint, and later diagnostic analyses. Later diagnostics reused fixed states, source bindings, splits, and outcomes; they were not used to select or modify the frozen endpoint. No model training, tuning, result selection, or canonical numerical claim was changed for the manuscript identity reframe.

The machine-readable visibility map assigns every canonical claim to main headline, main support, one main system diagnostic, or supplement-only use. Source-hash records bind every number to its immutable result artifact, and the supplement manifest independently records copied-file digests. External-data records are feasibility audits only and contain no external method-performance evaluation.

Scope remains the paired 276-specimen, six-domain CFRP data program; exact native-raster digital cost; the fixed predictor families, state representations, action hierarchy, and planners; and the registered CAI and normalized-RGB-MSE tasks. Physical path length, coupling, settling, hardware timing, industrial cost, and external empirical transfer are not established.